

Hands on 1

Create a Spring Web Project using Maven

 **spring** initializr

Project

☐ Gradle - Groovy

☐ Gradle - Kotlin

☒ Maven

Language

☒ Java

☐ Kotlin

☐ Groovy

Spring Boot

☐ 4.1.1 (SNAPSHOT)

☒ 4.1.0

☐ 4.0.8 (SNAPSHOT)

☐ 4.0.7

☐ 3.5.16

Project Metadata

Group

com.cognizant

Artifact

spring-learn

Package name

com.cognizant.spring-learn

Packaging

☒ Jar

☐ War

Dependencies

ADD DEPENDENCIES

No dependency selected

GENERATE

CTRL + G

EXPLORE

CTRL + SPACE

...

Project Explorer

- spring-learn
 - src/main/java
 - com.cognizant.spring_learn
 - Country.java
 - SpringLearnApplication.java
 - src/main/resources
 - application.properties
 - country.xml
 - date-format.xml
 - src/test/java
 - com.cognizant.spring_learn
 - SpringLearnApplicationTests.java
 - JRE System Library [JavaSE-17]
 - Maven Dependencies
 - src
 - main
 - java
 - com
 - cognizant
 - spring_learn
 - Country.java
 - SpringLearnApplication.java
 - resources
 - application.properties
 - country.xml
 - date-format.xml
 - test
 - target
 - HELP.md
 - mvnw
 - mvnw.cmd
 - pom.xml

SpringLearnApplica...

application.prope...

date-format.xml

country.xml

Country.java

SpringLearnApplica...

spring-learn/pom...

Dependency Hierarchy [test]

Filter:

Dependency Hierarchy

- spring-boot-starter: 4.1.0 [compile]
 - spring-boot-starter-logging: 4.1.0 [compile]
 - logback-classic: 1.5.34 [compile]
 - logback-core: 1.5.34 [compile]
 - spring-jcl: 2.0.18 (managed from 2.0.17) [compile]
 - spring-jcl: 2.0.18 (managed from 2.0.17) [compile]
- spring-boot-autoconfigure: 4.1.0 [compile]
 - spring-boot: 4.1.0 [compile]
 - spring-core: 7.0.8 [compile]
 - spring-context: 7.0.8 [compile]
 - spring-aop: 7.0.8 [compile]
 - spring-beans: 7.0.8 [compile]
- spring-expression: 7.0.8 [compile]
 - spring-core: 7.0.8 [compile]
- micrometer-observation: 1.17.0 (managed from 1.16.6) [compile]

Resolved Dependencies

- accessors-smart: 2.6.0 [test]
- android-json: 0.0.20131108.vaadin1 [test]
- apiguardian-api: 1.1.2 [test]
- asm: 9.7.1 [test]
- assertj-core: 3.27.7 [test]
- awaitility: 4.3.0 [test]
- byte-buddy: 1.18.10 [test]
- byte-buddy-agent: 1.18.10 [test]
- commons-logging: 1.3.6 [compile]
- hamcrest: 3.0 [test]
- jakarta.activation-api: 2.1.4 [test]
- jakarta.annotation-api: 3.0.0 [compile]
- jakarta.xml.bind-api: 4.0.5 [test]
- jsonassert: 1.5.3 [test]
- json-path: 2.10.0 [test]
- json-smart: 2.6.0 [test]
- jspecify: 1.0.0 [compile]
- jul-to-slf4j: 2.0.18 [compile]
- junit-jupiter: 6.0.3 [test]
- junit-jupiter-api: 6.0.3 [test]
- junit-jupiter-engine: 6.0.3 [test]
- junit-jupiter-params: 6.0.3 [test]

Problems

Javadoc

Declaration

Console

```
<terminated> SpringLearnApplication [Java Application] C:\Users\anju\AppData\Local\Programs\Eclipse Adoptium\jdk-17.0.9-hotspot\bin\javaw.exe (09-Jul-2026, 5:13:19 pm - 5:13:20 pm el
setCountry()Inside setCountry
<init>Inside Country Constructor.
setCode()Inside setCode()
setName()Inside setName()
<init>Inside Country Constructor.
setCode()Inside setCode()
```

Hands on 2

Spring Core – Load SimpleDateFormat from Spring Configuration XML

SimpleDateFormat with the pattern 'dd/MM/yyyy' is created in multiple places of an application. To avoid creation of SimpleDateFormat in multiple places, define a bean in Spring XML Configuration file and retrieve the date.

The screenshot displays the Eclipse IDE interface with two main panels. The left panel shows the Project Explorer with a project named 'spring-learn' containing a 'resources' folder with 'application.properties' and 'date-format.xml'. The right panel is split into two sections: the top section shows the 'date-format.xml' file, and the bottom section shows the 'SpringLearnApplication.java' file.

date-format.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="
         http://www.springframework.org/schema/beans
         https://www.springframework.org/schema/beans/spring-beans.xsd">
    <bean id="dateFormat"
          class="java.text.SimpleDateFormat">
        <constructor-arg value="dd/MM/yyyy"/>
    </bean>
</beans>
```

SpringLearnApplication.java

```
package com.cognizant.spring_learn;
import java.text.SimpleDateFormat;
import java.util.Date;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.annotation.*;
import org.springframework.context.support.ClassPathXmlApplicationContext;

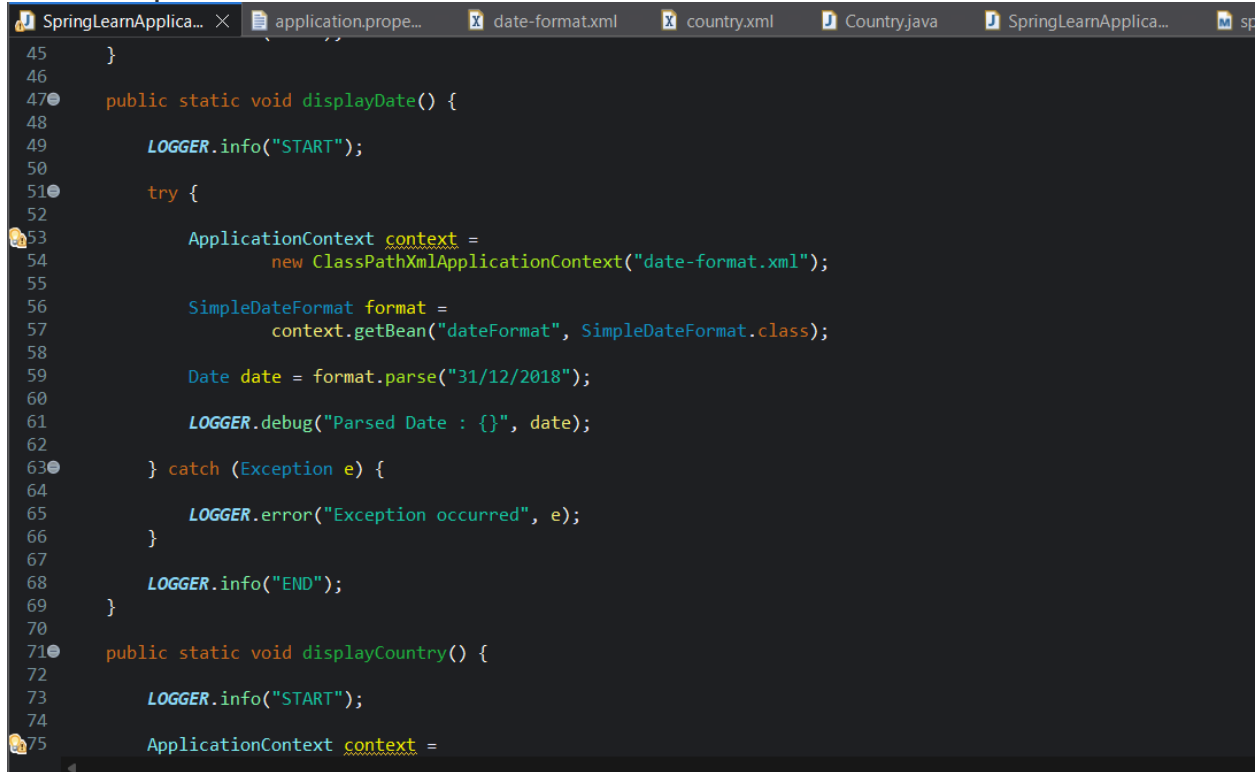
@SpringBootApplication
public class SpringLearnApplication {
    private static final Logger LOGGER =
        LoggerFactory.getLogger(SpringLearnApplication.class);
}
```

The bottom panel shows the console output of the application. It starts with the Spring Boot logo and the text 'Spring Boot (v4.1.0)'. The output shows the application starting successfully and displaying the date 'Mon Dec 31 00:00:00 IST 2018'.

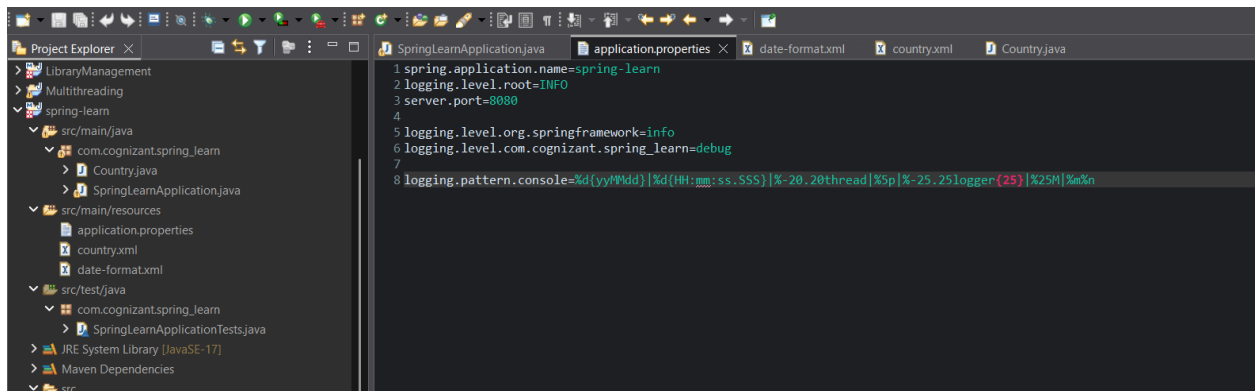
Hands on 3

Spring Core - Incorporate Logging

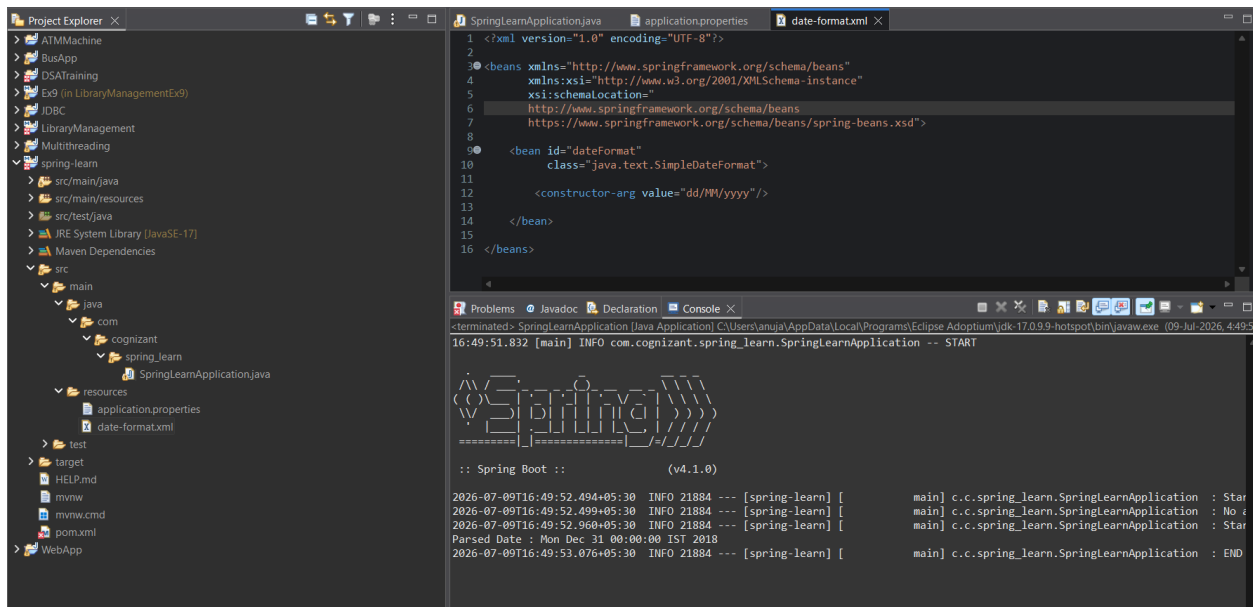
Incorporate logging in the Spring Boot project created in previous hands on. Refer steps below:



```
45     }
46
47     public static void displayDate() {
48
49         LOGGER.info("START");
50
51         try {
52
53             ApplicationContext context =
54                 new ClassPathXmlApplicationContext("date-format.xml");
55
56             SimpleDateFormat format =
57                 context.getBean("dateFormat", SimpleDateFormat.class);
58
59             Date date = format.parse("31/12/2018");
60
61             LOGGER.debug("Parsed Date : {}", date);
62
63         } catch (Exception e) {
64
65             LOGGER.error("Exception occurred", e);
66
67         }
68
69         LOGGER.info("END");
70     }
71
72     public static void displayCountry() {
73
74         LOGGER.info("START");
75
76         ApplicationContext context =
```



```
1 spring.application.name=spring-learn
2 logging.level.root=INFO
3 server.port=8080
4
5 logging.level.org.springframework=info
6 logging.level.com.cognizant.spring_learn=debug
7
8 logging.pattern.console=%d{yyMMdd} [%d{HH:mm:ss.SSS}] %-20.20thread [%5p] %-25.25logger{25} %25m|%n
```



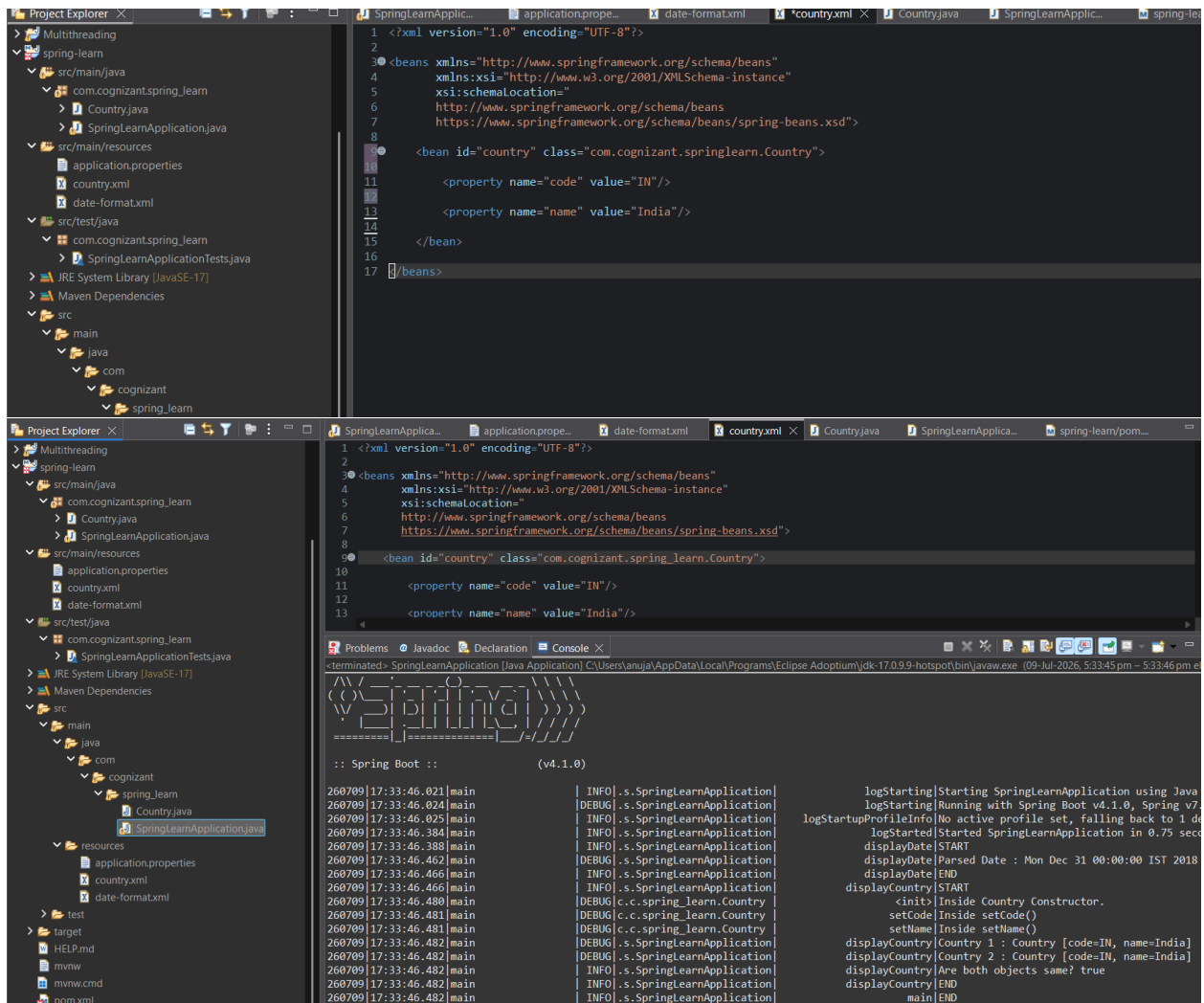
Hands on 4

Spring Core – Load Country from Spring Configuration XML

An airlines website is going to support booking on four countries. There will be a drop down on the home page of this website to select the respective country. It is also important to store the two-character ISO code of each country.

| Code | Name |
|------|---------------|
| US | United States |
| DE | Germany |
| IN | India |
| JP | Japan |

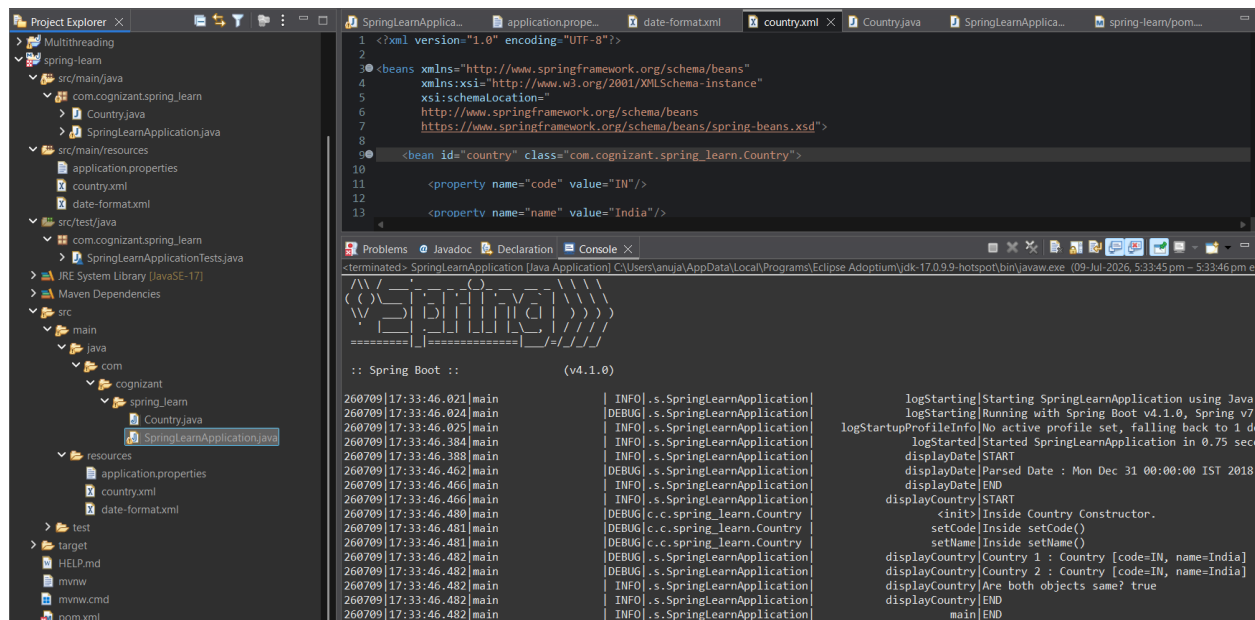
Above data has to be stored in spring configuration file. Write a program to read this configuration file and display the details.



Hands on 5

Spring Core – Demonstration of Singleton Scope and Prototype Scope

The Country bean done in the previous hands on will be used to demonstrate the scopes in Spring. Implement the steps below.



Hands on 6

Spring Core – Load list of countries from Spring Configuration XML

Our main objective was to retrieve the list of four countries for the airlines website.

